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From Chaotic Maps to SIMD-Native Image Transforms: A Stage-Level Performance and Security Evaluation

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ABSTRACT Chaos-based image-encryption studies commonly evaluate complete schemes and image-domain statistics, making it difficult to identify which design choices actually determine encryption throughput and whether favorable image statistics correspond to meaningful cryptographic resistance. This work presents a reproducible C++17 benchmark that decomposes representative chaotic image-encryption methods into keystream generation, pixel permutation, and diffusion. Traditional components are compared with SIMD-oriented replacements on 12 deterministic synthetic images and the 24-image Kodak PhotoCD dataset. The evaluation contains 14,730 performance observations and 3,300 full-scheme security observations, with ten repetitions per configuration. Replacing chaotic sort permutation with a linear block permutation improves mean permutation throughput by 100.6–131.9 times. Replacing a global feedback chain with AVX2 multi-lane or tree diffusion improves mean diffusion throughput by approximately 1.8–2.1 times. The fastest benchmark-informed candidate, Checkerboard-CA-ARX, sustains 259.1 MiB/s on Kodak images and 200.0 MiB/s at 4096×4096, while ChaCha20 remains substantially faster. Security experiments show that favorable entropy, histogram, correlation, and key-sensitivity values do not imply resistance to differential or known-plaintext attacks. The candidate pipelines are therefore presented as performance templates rather than deployment-ready ciphers.

INDEX TERMS AVX2, cellular automata, chaotic image encryption, image cryptanalysis, permutation-diffusion, SIMD, stage-level benchmarking.

I. INTRODUCTION

THE permutation-diffusion architecture is a recurring structure in chaos-based image encryption. A permutation stage rearranges pixel positions, and a diffusion stage modifies pixel values using a key-dependent sequence. The architecture maps naturally to common image statistics: permutation reduces adjacent-pixel correlation, while diffusion can flatten histograms and increase sensitivity to key changes.

However, two problems complicate performance and security claims in this literature. First, complete-scheme timing hides the contribution of individual stages. A proposed SIMD diffusion kernel may be fast while an $O(N \log N)$ chaotic sort still dominates total execution time. Conversely, a sequential chaotic generator may prevent a pipeline from benefiting from an otherwise parallel permutation. Second, image-domain metrics such as entropy, correlation, NPCR, and UACI are diagnostic measurements rather than proofs of cryptographic security. A deterministic stream-XOR construction can produce a uniform-looking ciphertext while remaining vulnerable when a key/nonce pair is reused.

This study investigates how implementation-friendly stages affect the performance of chaotic image-encryption methods without conflating that engineering result with cryptographic security. Traditional chaotic components, SIMD-oriented alternatives, and standard cryptographic baselines are implemented within one benchmark suite. The experiments isolate keystream, permutation, and diffusion execution time before composing selected components into benchmark-informed candidate pipelines.

The study answers four research questions:

- **RQ1:** Which stages dominate representative chaotic image-encryption pipelines?
- **RQ2:** Which replaceable stage designs benefit most from linear-time and SIMD-oriented restructuring?
- **RQ3:** Do stage-level improvements remain visible in composed candidate pipelines and at larger image sizes?

- **RQ4:** Which security conclusions are, and are not, supported by common image-domain metrics?

The contributions are:

1) A reproducible C++17/OpenCV/OpenSSL benchmark with a common cipher interface and separate keystream, permutation, diffusion, and total timing.
2) A controlled comparison of traditional components and SIMD-oriented replacements across synthetic images, real natural images, and resolutions up to 4096×4096.
3) Evidence that linear block permutation and lane/tree diffusion remove major implementation bottlenecks, while scalar floating-point maps and sort-based permutation remain expensive.
4) A negative security result demonstrating that strong image statistics can coexist with differential and reused-nonce known-plaintext weaknesses.
5) An explicit boundary between fast image-domain transformation prototypes and cryptographically deployable encryption.

II. BACKGROUND AND RELATED WORK

A. PERMUTATION-DIFFUSION IMAGE ENCRYPTION

Shannon identified confusion and diffusion as fundamental properties of secret-key systems [1]. Fridrich adapted related ideas to image encryption using two-dimensional chaotic maps and a permutation-diffusion architecture [2]. Many later schemes use pixel-, bit-, or block-level permutation followed by one or more diffusion rounds [3], [4].

Traditional chaotic permutation often generates one chaotic score per pixel and sorts pixel indices by those scores. The method is simple to describe but requires $O(N \log N)$ comparisons and irregular memory access. Diffusion is frequently implemented as the global recurrence

$$C[i] = P[i] \oplus K[i] \oplus C[i-1], \tag{1}$$

where P, K, and C denote permuted plaintext, keystream, and ciphertext. This construction creates an apparent avalanche path but also introduces a loop-carried dependency that restricts vectorization.

B. CHAOTIC AND SPATIAL KEYSTREAM GENERATORS

Scalar logistic, tent, and sine maps are common keystream sources. Their iterative dependency and floating-point operations can limit throughput. Spatial systems such as coupled-map lattices and cellular automata expose multiple local states and are therefore more compatible with lane-level or data-parallel execution [5]. This work evaluates both categories, as well as Hamiltonian-inspired and reaction-diffusion generators.

C. SIMD-ORIENTED RESTRUCTURING

Single Instruction, Multiple Data (SIMD) instructions operate on multiple values per instruction. On the evaluated x86-64 host, AVX2 provides 256-bit integer vectors. SIMD benefits are largest when work can be expressed as regular independent operations. Sorting, pointer chasing, and global recurrences are less suitable.

Associative operations can sometimes be reformulated as scans. The global XOR chain can be written as

$$X[i] = P[i] \oplus K[i], \tag{2}$$

$$C[i] = IV \oplus X[0] \oplus \cdots \oplus X[i]. \tag{3}$$

This permits block-wise prefix processing with a carry between blocks. Multi-lane chaining instead maintains 32 independent byte chains per AVX2 vector:

$$C[i] = P[i] \oplus K[i] \oplus C[i-32]. \tag{4}$$

Tree diffusion applies shuffle-and-XOR stages to mix bytes within a vector in logarithmic depth. Parallel scan and SIMD prefix techniques are established general-purpose primitives [6].

D. SECURITY EVALUATION

Entropy, histogram uniformity, adjacent-pixel correlation, NPCR, and UACI are widely reported in image-encryption studies. NPCR and UACI quantify output changes caused by a small input modification [7]. These metrics are useful, but they do not replace cryptanalysis. Prior work has demonstrated chosen-plaintext and structural attacks against permutation-diffusion image ciphers [8], [9].

The present study therefore includes AES-256-CTR and ChaCha20 as standardized or widely deployed cryptographic reference points [10], [11]. BLAKE3 keyed extendable output is also evaluated as a high-quality keystream source [12]. These baselines anchor the performance discussion; the proposed stage combinations are not claimed to provide equivalent security.

III. BENCHMARK DESIGN

A. COMMON PIPELINE MODEL

Each complete scheme implements a common *IImageCipher* interface. Encryption is represented as three timed stages:

1) **Keystream generation:** produce N key-dependent bytes.
2) **Permutation:** rearrange pixels, bytes, blocks, or local positions.
3) **Diffusion:** combine the permuted data and keystream.

Total execution time includes stage execution and required data conversion or allocation overhead. Throughput, reported in mebibytes per second, is

$$\text{throughput} = \frac{\text{image bytes}}{\text{total execution time}}. \tag{5}$$

The suite also executes each replaceable stage independently. Checksums prevent unused outputs from being optimized away.

B. FULL-SCHEME BASELINES

The full-scheme suite contains scalar chaotic XOR methods using logistic, tent, sine, coupled-lattice, and Hamiltonian-lattice generators; permutation plus XOR methods using logistic sort, Arnold map, and tiled Arnold map; an official BLAKE3 keyed-XOF keystream followed by XOR; and AES-256-CTR and ChaCha20 through OpenSSL EVP.

All full schemes are reversible and checked by encryption followed by decryption. The benchmark intentionally reuses a fixed key and nonce across security probes to expose reused-keystream behavior. This is a misuse test, not a recommended operating mode for AES-CTR, ChaCha20, or BLAKE3.

C. REPLACEABLE STAGES

The isolated keystream experiments evaluate scalar logistic and sine maps, fixed-point tent, a 16-state coupled-map lattice (CML), a 64-cell rule-30-like cellular automaton (CA), a Hamiltonian-inspired lattice, and a reaction-diffusion generator. These generators compare computational structures and are not claimed to provide cryptographic unpredictability.

The permutation experiments compare chaotic score sort, sequential random walk, Arnold/cat map, affine modular mapping, key-derived Feistel-like mapping with cycle walking, 256-byte block permutation, and four rounds of disjoint checkerboard swaps. The block and checkerboard implementations are deliberately simple performance templates. They are fixed rather than key-derived and must not be interpreted as secure permutations.

The diffusion experiments include global and block-local XOR chains, blocked prefix processing, a two-dimensional stencil, ARX block and bit-plane-style transforms, forward and reverse prefix XOR, 32-lane AVX2 chaining, AVX2 tree XOR, a forward-prefix/tree/reverse-prefix composition, and ARX prefix modulo 256. The prefix-XOR prototype uses AVX2 loads, XORs, broadcasts, and stores, but its within-vector prefix helper performs a scalar scan. The multi-lane and tree variants contain the clearest AVX2-native data paths.

D. CANDIDATE PIPELINES

Seven candidate combinations test whether isolated improvements survive composition. Table 1 summarizes their stages.

These pipelines currently produce checksums for performance validation but do not implement complete decryptors or equivalent security probes. They are component-composition experiments, not proposed secure ciphers.

IV. EXPERIMENTAL METHODOLOGY

A. IMPLEMENTATION AND PLATFORM

The benchmark is implemented in C++17 and built with CMake in Release mode using GCC 11.4.0, `-O3`, `-march=native`, and explicit AVX2 support. OpenCV is used for image loading and output, OpenSSL EVP provides AES-256-CTR and ChaCha20, and the official BLAKE3 C implementation provides keyed XOF output.

Experiments ran on a four-core Intel Xeon E5-2620 v4 at 2.10 GHz under 64-bit Linux 5.15. The processor supports SSE2, AVX, AVX2, and AES instructions. BLAKE3 was compiled in portable mode, so its result does not include optional architecture-specific SIMD dispatch. Candidate and stage measurements are single-process measurements; this paper does not claim multicore scaling.

B. DATASETS

Two dataset classes are used:

- **Synthetic controls:** deterministic gradient, texture, and noise images at 512×512, 1024×1024, 2048×2048, and 4096×4096.
- **Natural images:** all 24 color images from the Kodak PhotoCD dataset [13], comprising 768×512 and 512×768 orientations.

Synthetic images provide controlled size and content variation. Kodak images provide natural-image diversity while retaining a stable public benchmark. The Lena/Lenna image is explicitly excluded from the publication dataset and does not appear in the active benchmark results.

C. EXPERIMENT MATRIX

Every executed configuration is repeated ten times. Slow full schemes and intentionally slow stage baselines are limited to images no larger than 1024×1024. Fast stage alternatives continue to 4096×4096. Slow candidate pipelines continue to 2048×2048, and fast candidates continue to 4096×4096.

The final dataset contains 3,300 full-scheme performance records, 3,300 full-scheme security records, 9,000 isolated-stage performance records, and 2,430 candidate-pipeline performance records. Reported aggregate throughput values are arithmetic means. The 95% confidence interval is

$$\mathrm{CI}_{95} = \frac{1.96s}{\sqrt{n}}, \tag{6}$$

where s is the sample standard deviation. No outlier removal is applied.

D. SECURITY METRICS

For full schemes, the suite records Shannon entropy, the 256-bin chi-square statistic, horizontal/vertical/diagonal adjacent-pixel correlation, NPCR and UACI after flipping one plaintext bit, NPCR after flipping one key bit, empirical known-plaintext and chosen-plaintext byte-recovery scores under a reused key/nonce, and exact decryption correctness.

For the byte-recovery probe, a keystream estimate is computed as $K'[i] = P_1[i] \oplus C_1[i]$ and applied to a second ciphertext. A score of 1.0 means complete recovery under this simple reused-keystream model.

V. RESULTS

A. FULL-SCHEME PERFORMANCE

Table 2 reports selected aggregate full-scheme results. ChaCha20 is the fastest complete baseline on all commonly measured image dimensions. AES-256-CTR is second among the standard baselines. BLAKE3-XOR is the fastest non-OpenSSL full scheme, but its portable-only BLAKE3 build makes this a conservative measurement.

The results do not support a claim that chaotic-image pipelines outperform standard ciphers. Instead, they show the

TABLE 1. Benchmark-Informed Candidate Pipelines

Candidate	Keystream	Permutation	Diffusion
CA-Feistel-ARX	Cellular automata	Feistel-like index	ARX block
Checkerboard-CA-ARX	Cellular automata	Checkerboard swaps	ARX block
Affine-CA-PrefixTree	Cellular automata	Affine mapping	Prefix/tree/reverse
Checkerboard-CA-MultilaneTree	Cellular automata	Checkerboard swaps	Multi-lane/tree
CML-Feistel-Stencil	CML	Feistel-like index	Stencil
Hamiltonian-Block-Stencil	Hamiltonian lattice	Block-Feistel mapping	Stencil
Affine-CML-Bitplane	CML	Affine mapping	Bit-plane transform

TABLE 2. Selected Aggregate Full-Scheme Throughput

Scheme	Image size	n	Mean MiB/s	95% CI	Speedup
ChaCha20	512×512	30	996.357	42.849	7.128×
ChaCha20	768×512	180	895.707	10.484	6.389×
ChaCha20	1024×1024	30	903.400	18.640	6.532×
AES-256-CTR	768×512	180	355.663	3.399	2.537×
AES-256-CTR	1024×1024	30	594.958	9.202	4.302×
BLAKE3-XOR	768×512	180	315.714	2.750	2.252×
BLAKE3-XOR	1024×1024	30	294.142	4.023	2.127×

TABLE 3. Block-Permutation Performance Relative to Chaotic Sort

Size	n	MiB/s	CI	Speedup
512×512	30	512.371	6.473	100.634×
512×768	60	499.309	4.729	110.013×
768×512	180	499.812	2.866	109.371×
1024×1024	30	497.839	5.289	131.861×

cost of common chaotic components and provide a performance target for alternative stage designs.

B. ISOLATED PERMUTATION PERFORMANCE

Permutation produces the largest stage-level improvement. At 512×512, block permutation reaches 512.371 MiB/s and is 100.634 times faster than chaotic sort. At 1024×1024 it reaches 497.839 MiB/s and is 131.861 times faster.

This result answers RQ1 and RQ2: chaotic sort is implementation-hostile, and replacing it with a regular linear-time transform changes the performance regime. The benchmark block permutation is fixed and must be made key-derived and cryptographically analyzed before use in a cipher.

C. ISOLATED DIFFUSION PERFORMANCE

The fastest diffusion alternatives are multi-lane chaining and tree XOR. Their mean speedups over the global chain are approximately twofold.

Removing a global byte-to-byte dependency creates a measurable SIMD opportunity. It also explains why adding SIMD only to XOR inside a traditional end-to-end pipeline may show little total improvement: the remaining generator and permutation costs dominate.

D. CANDIDATE PIPELINE PERFORMANCE

Checkerboard-CA-ARX is the fastest composed candidate. On Kodak images it achieves 259.138 MiB/s, while Checkerboard-CA-MultilaneTree achieves 232.142 MiB/s. Both remain slower than AES-256-CTR and ChaCha20 on the same image dimensions.

Table 6 shows that the fast checkerboard candidates distribute work across all three stages, whereas Feistel indexing dominates CA-Feistel-ARX and Hamiltonian generation dominates Hamiltonian-Block-Stencil. These results support RQ3: isolated-stage choices remain visible after composition.

E. FULL-SCHEME SECURITY DIAGNOSTICS

All complete schemes decrypt correctly. Several also produce near-eight-bit entropy, low adjacent-pixel correlation, and strong key sensitivity. These positive metrics are insufficient to establish security.

The KPA recovery score of 1.0 for AES-CTR, ChaCha20, BLAKE3-XOR, and the stream-XOR chaotic variants is expected because the experiment deliberately reuses the same key/nonce. It demonstrates misuse sensitivity, not a break of the underlying standardized primitive. The negligible plaintext NPCR is also expected for CTR- or stream-XOR encryption: changing one plaintext bit changes only the corresponding ciphertext bit when the keystream is unchanged.

This result answers RQ4. Entropy, correlation, and key sensitivity can look excellent while a construction still lacks plaintext avalanche and fails under reused keystream. NPCR is not an appropriate requirement for all secure encryption modes; standard stream ciphers intentionally do not provide plaintext avalanche. The candidate pipelines were not subjected to the same security harness and contain fixed prototype permutations. No cryptographic-security conclusion is made for them.

VI. DISCUSSION

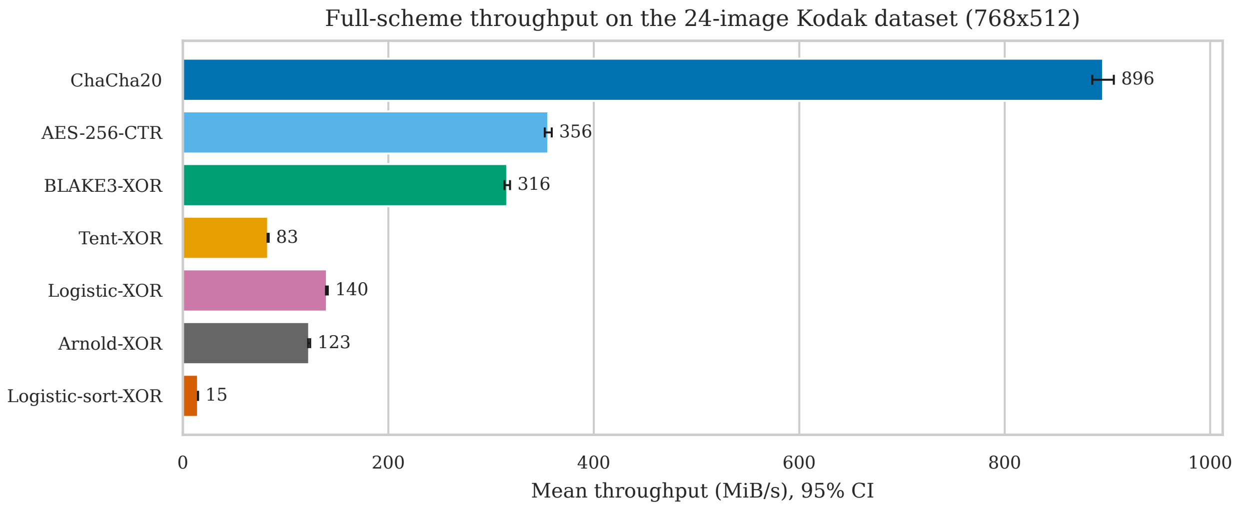

FIGURE 1. Selected full-scheme throughput on the 24-image Kodak dataset. Error bars show 95% confidence intervals. Standard cryptographic baselines remain substantially faster than the evaluated chaotic full schemes.

TABLE 4. Selected Aggregate Diffusion Performance

Stage	Image size	n	Mean MiB/s	95% CI	Speedup
Multi-lane chain AVX2	512×512	30	499.419	7.774	2.076×
Multi-lane chain AVX2	768×512	180	499.120	3.046	2.095×
Multi-lane chain AVX2	1024×1024	30	493.067	8.000	2.054×
Tree XOR AVX2	512×512	30	470.202	7.667	1.955×
Tree XOR AVX2	768×512	180	469.901	2.793	1.973×
Tree XOR AVX2	1024×1024	30	464.386	8.035	1.935×
ARX block diffusion	768×512	180	426.568	2.540	1.791×

TABLE 5. Aggregate Candidate-Pipeline Throughput

Candidate	Image size	n	Mean MiB/s	95% CI
Checkerboard-CA-ARX	512×512	30	267.596	3.771
Checkerboard-CA-ARX	768×512	180	259.138	2.666
Checkerboard-CA-ARX	1024×1024	30	234.149	4.490
Checkerboard-CA-ARX	2048×2048	30	216.487	3.933
Checkerboard-CA-ARX	4096×4096	30	199.993	4.384
Checkerboard-CA-MultilaneTree	768×512	180	232.142	2.111
Checkerboard-CA-MultilaneTree	4096×4096	30	159.392	3.931

TABLE 6. Mean Candidate Stage Shares

Candidate	Key	Perm.	Diff.
Checkerboard-CA-ARX	47.69%	27.86%	24.44%
Checkerboard-CA-MultilaneTree	38.44%	22.60%	38.95%
CA-Feistel-ARX	11.69%	82.32%	5.99%
Hamiltonian-Block-Stencil	83.78%	11.97%	4.24%

A. WHAT SIMD CHANGES

The experiments show that SIMD is not a property of a pipeline label. It is a property of the dependency graph and memory-access pattern of each stage. Regular, independent byte or word operations benefit from AVX2. Global recurrences, transcendental scalar maps, cycle walking, and sort-based permutation restrict the gain.

The largest improvement does not come from a vector instruction alone. It comes from replacing chaotic sort with a linear regular mapping. Similarly, the diffusion improvement comes from changing one global chain into multiple lanes or a tree. Algorithmic restructuring precedes vectorization.

B. PERFORMANCE VERSUS SECURITY

The fastest stage is not necessarily the strongest stage. Fixed block and checkerboard permutations are useful controls for measuring the cost of regular memory movement, but they provide no secret permutation by themselves. The CA generator is fast, but its rule-30-like output has not been established as a cryptographically secure pseudorandom stream. Tree XOR and linear multi-lane diffusion are also structurally analyzable.

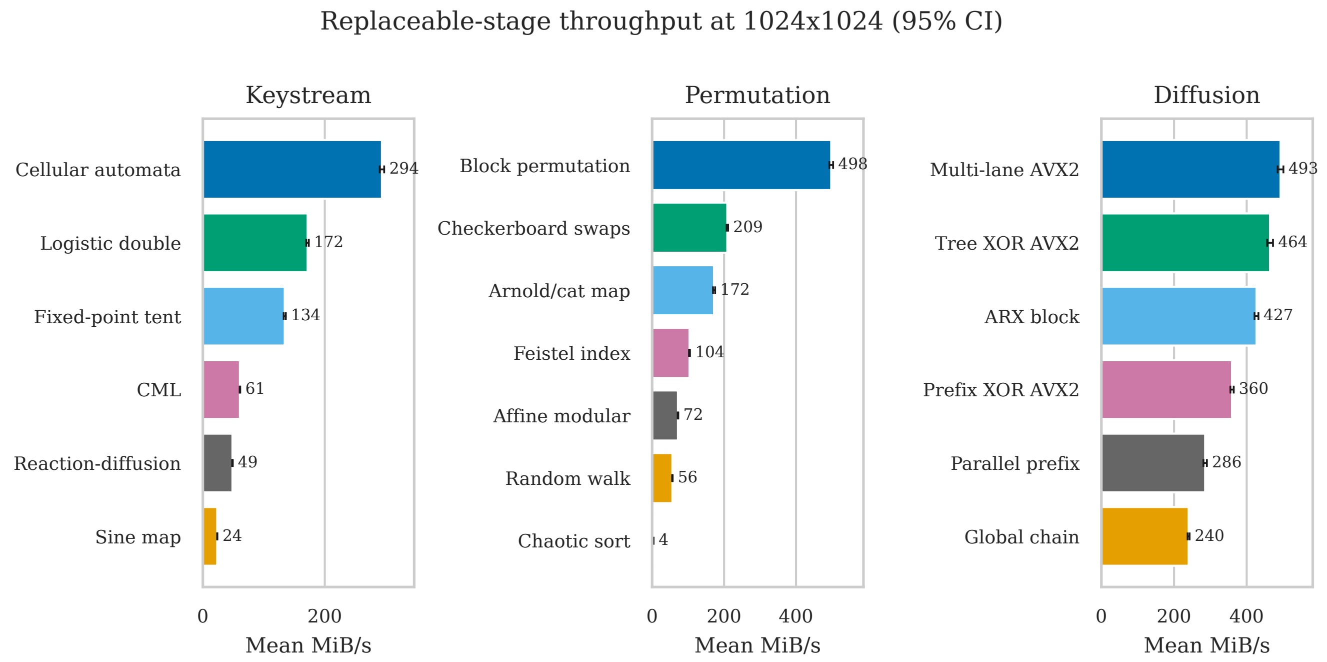

FIGURE 2. Representative replaceable-stage throughput at 1024 × 1024. The largest gain is algorithmic: replacing chaotic sort with a regular linear-time permutation. AVX2-native multi-lane and tree diffusion also outperform the sequential global chain.

TABLE 7. Mean Security Diagnostics Across 300 Observations per Scheme

Scheme	Entropy	χ^2	Abs. corr.	NPCR	Key sens.	KPA
ChaCha20	7.999845	258.874	0.001216	0.000084%	99.619%	1.000
AES-256-CTR	7.999843	259.125	0.001148	0.000084%	99.611%	1.000
BLAKE3-XOR	7.999850	247.607	0.001419	0.000084%	99.607%	1.000
Tent-XOR	7.999828	282.731	0.001721	0.000084%	99.623%	1.000
Logistic-XOR	7.984321	26641.399	0.020440	0.000084%	99.387%	1.000
Coupled-lattice-XOR	7.971770	45857.479	0.137337	0.000084%	99.410%	1.000

A deployment-oriented design should use a standard cryptographic primitive to derive independent subkeys and masks, make every permutation key-derived, include unique nonces, and provide authentication. A practical research direction is to retain image-domain transforms only as a format-aware preprocessing or selective-encryption layer, then protect the result with an authenticated standard cipher.

C. IMPLICATIONS FOR FUTURE STUDIES

Future evaluations should report isolated stage execution time; include AES and ChaCha through optimized libraries; compare algorithmic complexity before attributing speedup to SIMD; test real images and controlled synthetic inputs; report repeated measurements with confidence intervals; separate image statistics from cryptographic claims; and include negative controls for nonce reuse, known plaintext, and chosen plaintext.

VII. THREATS TO VALIDITY AND LIMITATIONS

Measurements were collected on one older x86-64 server CPU and may differ on newer AVX2/AVX-512 processors, ARM NEON/SVE systems, GPUs, or embedded devices. CPU frequency, virtualization noise, memory allocation, and cache state were not controlled through hardware performance counters or processor pinning. The benchmark reports wall-clock stage time and throughput, but not cycles per byte, energy, cache misses, branch misses, or instruction counts.

The prefix-XOR helper is only partially vectorized because its intra-vector scan is scalar. BLAKE3 is compiled without architecture-specific SIMD dispatch. Fixed block and checkerboard permutations are performance templates, not secure key-derived permutations. Candidate pipelines do not yet implement inverse transforms or undergo the full cryptanalysis suite. Image-domain diagnostics are reported only for full schemes and do not constitute proofs of security. AES-CTR and ChaCha20 are confidentiality-only baselines; deployment should use authenticated encryption.

These limitations constrain the claim to stage-level performance analysis and prototype composition. They do not invalidate the finding that sort permutation and global dependencies are major implementation bottlenecks.

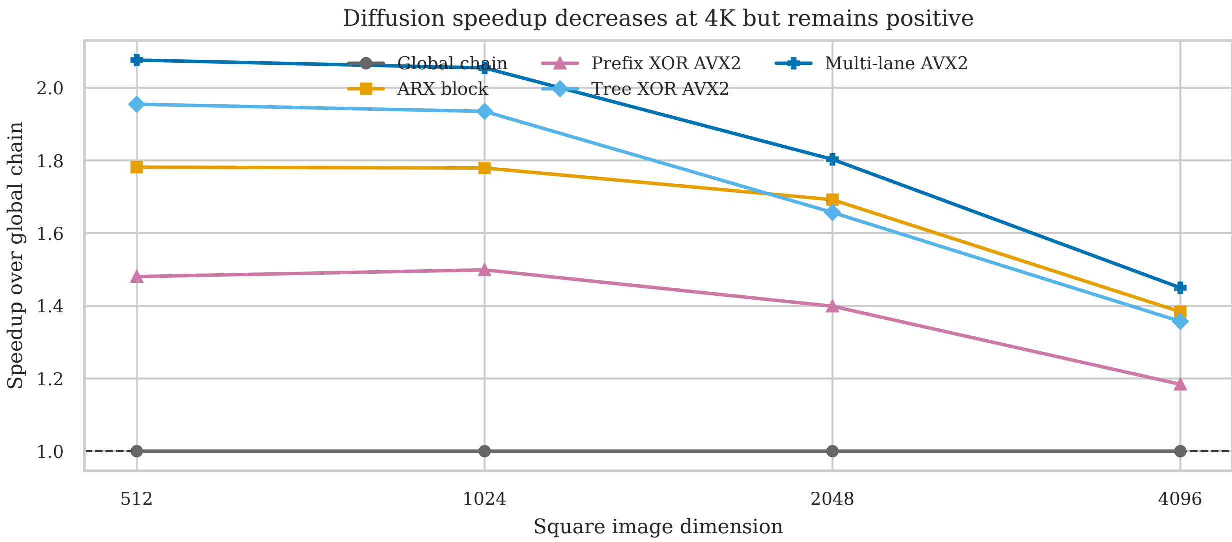

FIGURE 3. Diffusion speedup relative to the global chain on square images. AVX2 multi-lane and tree diffusion retain a positive advantage at all measured sizes, although the advantage narrows at 4096×4096.

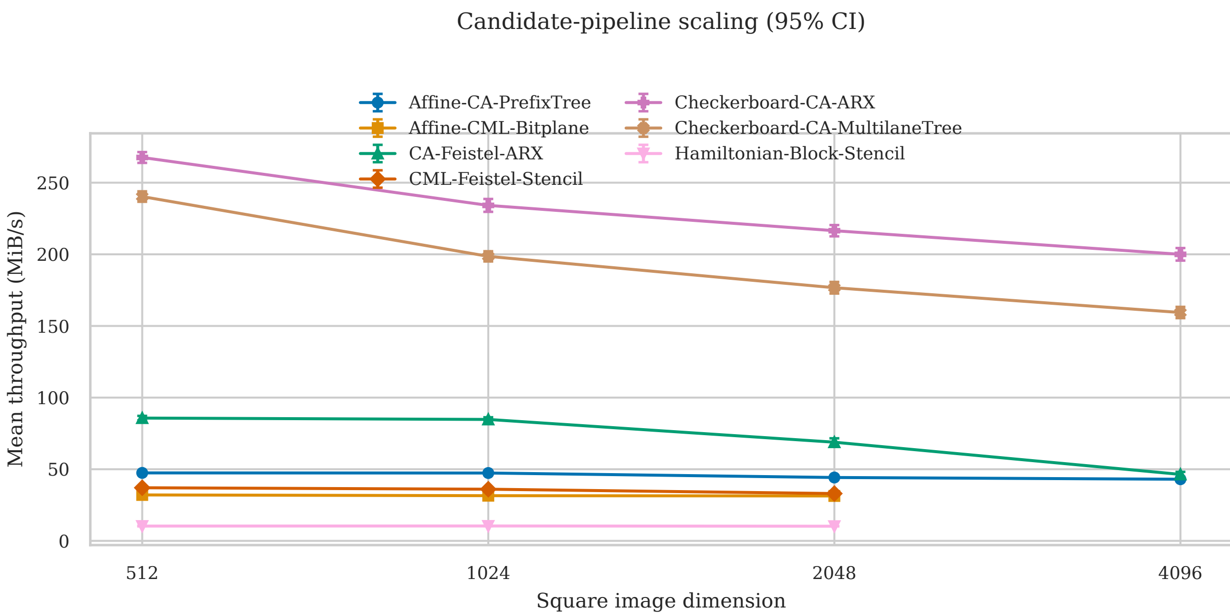

FIGURE 4. Candidate-pipeline throughput across square image sizes. The checkerboard candidates remain fastest, but all candidates lose throughput as working sets grow. Error bars show 95% confidence intervals.

VIII. REPRODUCIBILITY

The public research artifact is maintained at https://github.com/ravikanthreddy89/chaotic-encryption-survey. The repository includes the benchmark implementation, dataset generators, Kodak downloader, experiment harness, raw outputs, aggregate CSV files, figure-generation scripts, and manuscript source. A clean checkout can be built and the final experiment reproduced with:

```
git clone <repository-url>
cd chaotic-encryption-survey
cmake -S . -B build \
  -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
REPS=10 REAL_REPS=10 \
SIZES="512 1024 2048 4096" \
./scripts/run_paper_ready_experiments.sh
```

Primary result files include `results/final/bench.csv`, `stage_bench.csv`, `candidate_schemes.csv`, `analysis.csv`, `dataset_manifest.csv`, and their aggregate statistics. The repository README docu-

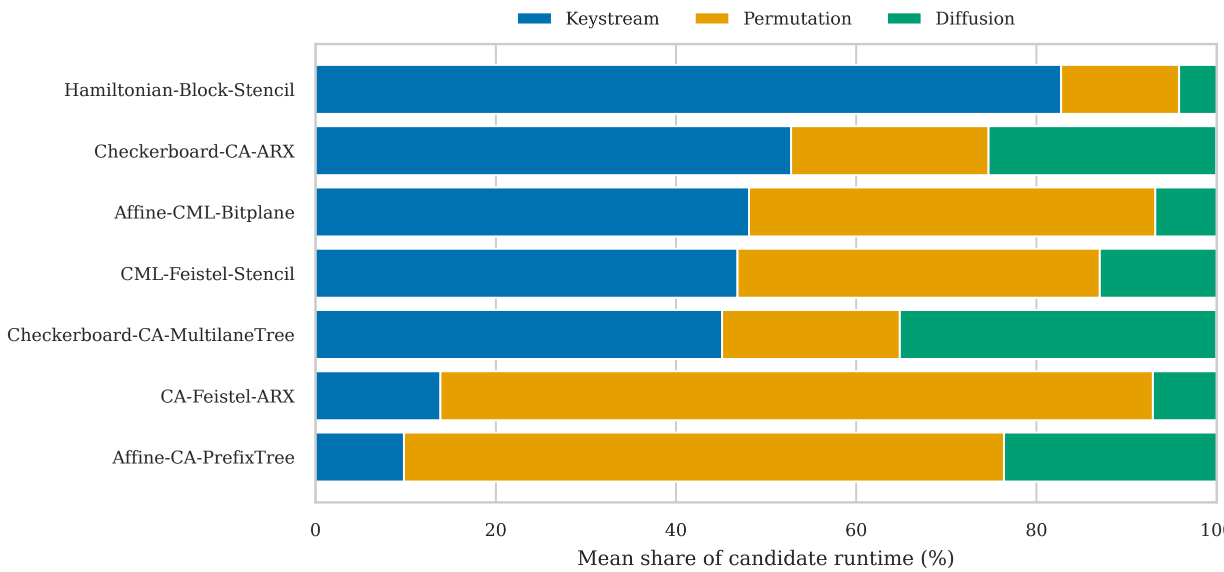

FIGURE 5. Mean runtime composition of the candidate pipelines. The plot exposes the dominant bottleneck in each composition and shows why isolated stage measurements remain necessary.

ments dependencies, quick-start commands, plot generation, and manuscript compilation. The complete source revision used by each experiment is retained in the generated metadata.

IX. CONCLUSION

This work decomposed chaotic image-encryption pipelines into replaceable keystream, permutation, and diffusion stages and evaluated traditional and SIMD-oriented alternatives. Implementation-hostile components, especially chaotic sort and global feedback dependencies, determine encryption throughput more strongly than the mere use of SIMD instructions.

Linear block permutation improves throughput by more than two orders of magnitude relative to chaotic sort. AVX2 multi-lane and tree diffusion approximately double throughput relative to a global chain. These improvements remain visible in composed candidate pipelines and at resolutions up to 4096×4096. Nevertheless, optimized ChaCha20 and AES-256-CTR remain faster and have substantially stronger security foundations.

The security experiments reinforce an equally important conclusion: image-domain statistics must not be presented as cryptographic proof. The current XOR-based full schemes show favorable entropy and key sensitivity but fail differential-avalanche and reused-keystream recovery controls. The benchmark-informed candidates are therefore performance prototypes, not secure ciphers.

DECLARATIONS

AUTHOR CONTRIBUTIONS

Ravikanth Reddy Gudipati conceived the study, implemented the benchmark, conducted the experiments, analyzed the results, and prepared the manuscript.

DATA AND CODE AVAILABILITY

The benchmark source code, experiment scripts, raw measurements, aggregate tables, manuscript source, and reproduction instructions are available at https://github.com/ravikanthreddy89/chaotic-encryption-survey.

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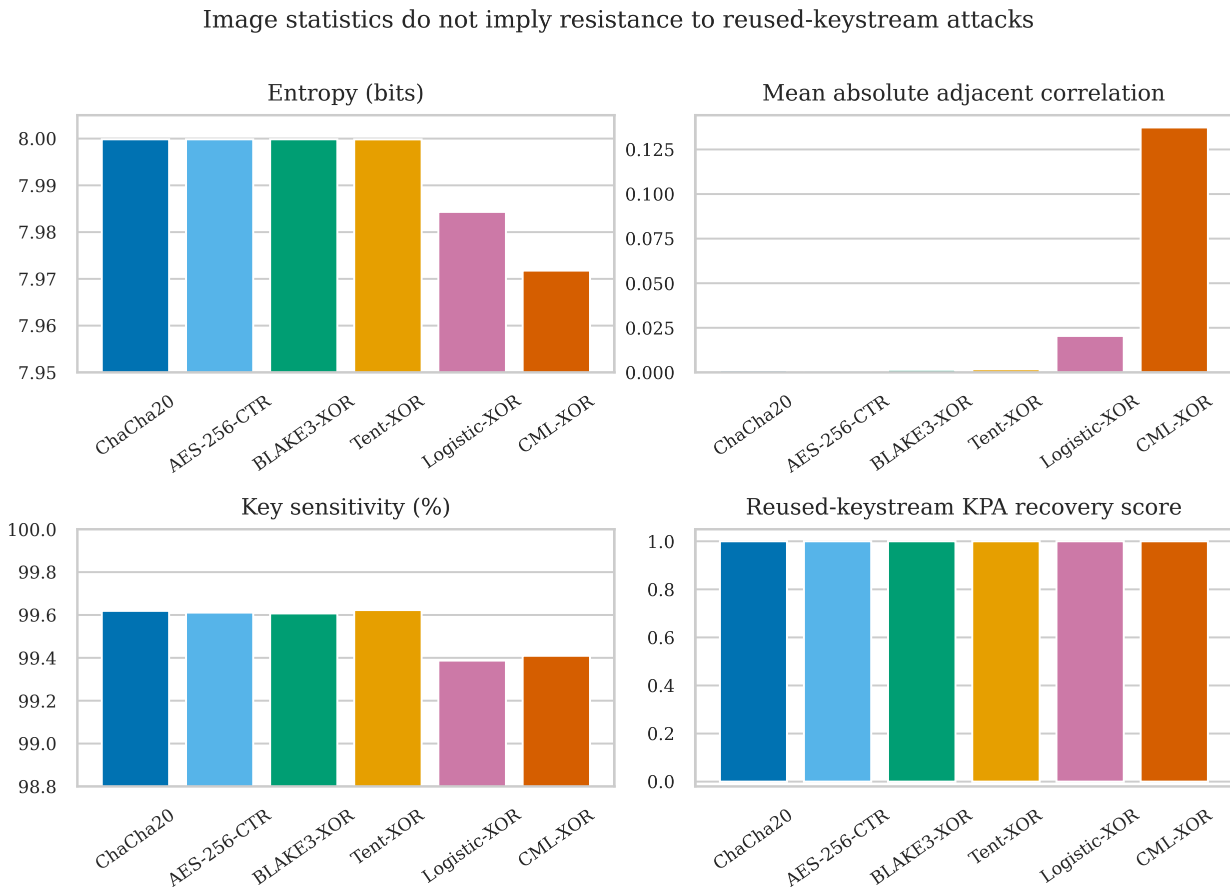

FIGURE 6. **Selected security diagnostics averaged over the full-scheme experiments. Near-ideal entropy, correlation, and key sensitivity coexist with complete recovery in the deliberately reused-keystream known-plaintext probe.**

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